

Week 2: Key Concepts and Terms

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This deck collects the **key concepts and terms** from Week 2: Introduction to Python.

For each term you should be able to:

- Define it in your own words
- Give an example or illustration
- Explain why it matters for your dissertation workflow

Use these slides as prompts — prepare your own definitions, answers and materials.

Note: This is a study aid only. It does not attempt to be comprehensive and should not be relied upon as sufficient preparation for assessment purposes.



Tools and Workflow

Term	What to know
Positron	What is it (editor + console + terminal)? What do you use it for in this module?
Editor	What is the editor used for vs the Console?
Console (REPL)	What does >>> mean? Why is it useful for quick experimentation?
Terminal (shell)	What does \$ / PS> mean? What kinds of commands belong here?

Term	What to know
Project folder	Why do we keep code, data, and outputs in one folder?
.py file	What is a Python script? How do you run it in Positron?
Interpreter session	What is a Python session, and what happens when you restart it?
Working directory	Why does file location matter when you run code?



Environments and Packages

Term	What to know
Virtual environment (<code>.venv</code>)	What problem does it solve? Why is it a "private toolbox" for a project?
Dependency	What is a dependency in a Python project?
Package	What is a package (e.g. <code>polars</code>)?
Import	What does <code>import polars as pl</code> do?

Term	What to know
<code>uv</code>	What does it manage (Python + packages)? Why do we use it in this module?
<code>uv venv</code>	What does it create and where?
<code>uv init</code>	What does it create in your project folder?
<code>uv add <pkg></code>	What changes when you install a package? Where do you run it (Terminal vs Console)?
<code>pyproject.toml</code>	What information is stored here? ("shopping list")
<code>uv.lock</code>	Why do we lock exact versions? (reproducibility)

Term	What to know
ModuleNotFoundError	What causes it? What is the standard fix?
Wrong tab	What goes wrong if you type Python into the Terminal or <code>uv add</code> into the Console?
Restart	When should you restart a session/terminal, and what does restarting do?



Core Python Concepts

Term	What to know
Expression	What is an expression (e.g. <code>2 + 2</code>) and what does it evaluate to?
Variable	What does it mean to store a value under a name?
Assignment (=)	What happens when you do <code>price = 142.5</code> ?
Function call	What is a function call (e.g. <code>print(...)</code>), and why use it?
Comment (#)	What is a comment for?

Term	What to know
<code>int / float</code>	What is the difference? Why does division often give floats?
<code>str</code>	What is a string and why must it be in quotes?
<code>bool</code>	What are <code>True</code> / <code>False</code> used for?
<code>type(x)</code>	What does it tell you?
<code>len(x) / sum(x) / max(x) / min(x)</code>	What kinds of inputs do these take?



Term	What to know
<code>print()</code>	When should you print instead of relying on the Console's last value?
f-string	What does the leading <code>f</code> do? How do braces <code>{ }</code> work?
Format specifier	What do you get from <code>:.2f</code> and why is it useful for finance outputs?

Data Structures

Term	What to know
List	What is a list and how do you create one with <code>[]</code> ?
Indexing	Why does indexing start at 0? What does <code>portfolio[0]</code> mean?
Negative index	What does <code>portfolio[-1]</code> mean?
<code>append</code>	What does <code>portfolio.append("GOOGL")</code> do?
<code>remove</code>	When does <code>remove</code> fail? How would you diagnose it?



Term	What to know
Dictionary	What is a dictionary and how do you create one with <code>{ }</code> ?
Key / value	What is the key and what is the value in <code>"price": 142.5</code> ?
Lookup	What does <code>stock["price"]</code> do?
Update	What happens when you do <code>stock["price"] = 145.0</code> ?
<code>items()</code>	What does iterating over <code>portfolio.items()</code> return?



Loops and Control Flow

Term	What to know
<code>for</code> loop	How do you read <code>for ticker in tickers:?</code> What repeats?
Indentation	Why does indentation matter in Python?
Iterable	What kinds of things can you loop over? (lists, dicts, <code>range</code>)
Accumulator	What pattern is <code>total += position</code> implementing?
<code>range()</code>	What does <code>range(5)</code> vs <code>range(1, 11)</code> produce?



while and Nested Loops

Term	What to know
while loop	When do you choose while instead of for ?
Condition	What kind of expression goes after while ?
Infinite loop	What causes infinite loops? How would you stop one?
Nested loop	What is a loop inside a loop used for?
Looping over dicts	What do you get from <code>for k, v in d.items():</code> ?



Summary Checklist

Can you define and give an example for each?

- Positron
- Editor
- Console (REPL)
- Terminal (shell)
- `.py` file
- Interpreter session
- Virtual environment (`.venv`)
- Package / dependency
- `uv venv` / `uv init`
- `uv add`
- `pyproject.toml`
- `uv.lock`
- Variable / assignment
- `int`, `float`, `str`, `bool`
- List / dictionary
- Indexing (0-based)
- `for` / `while`
- `range()`



Can you answer these in your own words?

- 1 What is the difference between the Console and the Terminal in Positron?
- 2 Why do we use a virtual environment for each project?
- 3 What files do `uv init` and `uv add` create or update?
- 4 What does `import ... as ...` do and why use an alias?
- 5 How do lists and dictionaries differ, and when would you choose each?
- 6 What does it mean that Python uses 0-based indexing?
- 7 Explain how a `for` loop works, including what indentation controls.
- 8 When would you use a `while` loop instead of a `for` loop?

